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Ensuring quality in software requirement engineering process: A comparative study

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ABSTRACT

Poorly managed requirements can lead to a software failure; hence, qualitative Requirement Engineering (RE) is essential for the success of software. Researchers have pointed out numerous ways to aid the RE process. This study aims to find practices deemed fit and unfit for quality improvement of the Software Requirement Engineering (SRE) process from the literature and verify them by software professionals. The study selects 57 articles published in journals and conferences of IEEE Xplore, ScienceDirect, and ACM Digital Library, from 2018 to 2023 to answer three research questions, yielding 8 quality practices. An industrial survey is then formulated to find the trends against those practices from the software industry. The findings from the literature and industrial survey are then compared. The comparison between literature and professional views proved ambiguous requirements to be the top cause of prolonged analysis and project failure. Also, requirement elicitation and analysis are the toughest RE activities. The quality practices pointed out by the literature make a positive difference in the quality of the developmental process of software and, if not followed, result in poorly managed or low-quality software products. Insufficient investment of time in engineering requirements can lead to cost and budget overruns, ultimately culminating in software failure.

1. Introduction

A software product's success depends on the people responsible for creating the software and the process followed for creating the software [1]. Requirement engineering (RE) serves as the fundamental stage in software engineering, encompassing four primary phases: elicitation, analysis and design, specification, and validation, alongside the overarching activity of management. It has been observed that inadequate time allocation to requirement engineering leads to cost and budget overruns, ultimately resulting in software failure [2,3,4,5,6]. Numerous practitioners [7,8,9,10,11,12] have defined and validated a range of steps and practices for each RE phase, enhancing its classification and comprehensiveness. These practices, identified by Karl and Joy as 'Good Practices' [13], are vital in aligning requirements with user needs and market demands, emphasizing quality [14]. While these practices do not guarantee project success, they significantly enhance the likelihood of successful project completion.

The research is focused on gaining insights into practices recommended or not recommended for the quality of the RE process and any tools that can help professionals achieve better quality. It gathers information presented by researchers from 2018 to 2023 about practices linked to the quality of the RE process. Also, the study lists any tools used or developed by researchers to achieve the quality. Based on the literature findings about quality hindrance and quality-enhancing practices,

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the study develops a questionnaire for industrial survey of the software industry. The survey results are compared against the literature findings to see how many practices co-exist between literature reportings and industrial practices. It addresses the following three questions:

RQ1: What mistakes compromise the quality of the RE process?

RQ2: Which practices are suggested for an improved RE process?

RQ3: Which tools and techniques help to improve the RE process?

This study aims to help organizations gauge their current RE processes against the theoretically reported practices and improve them to get top-quality software products.

The remainder of the article is organized as follows: Section II gives a background of basic concepts around quality-enhancing and quality hindering RE practices in the software industry. Section III explains the methodology devised for conducting this research. The findings of the study are reported in Section IV and the comparison between literature and industrial survey is discussed in Section V. The article closes with conclusion section, limitations, and future work.

2. Background

Software projects can fail due to a number of reasons, such as unfeasible objectives, vague requirements, informal recording of important data, improbable schedules, erroneous resource estimates, and insufficient assessment of risks [15,16]. At the beginning of the RE cycle, selection of an appropriate technique for gathering requirements is a crucial task as no single technique can cover all essential requirements. Various dynamics, such as knowledge of existing sources and accessibility of new domain knowledge sources, come into play while deciding on techniques best suited for a software project [17]. Problems, such as unclear project vision, ambiguous requirements, skipped stakeholders, and ineffective communication, can arise [18]. Therefore, it is vital for the requirement analyst to have a comprehensive understanding of various elicitation techniques in order to create the most effective combinations.

In the study conducted by Yousaf and Asger [19], elicitation techniques were categorized into four types, each suitable for a different stage of elicitation:

1. Traditional techniques: such as interviews and surveys
2. Contextual techniques: such as observations
3. Collaborative techniques: such as prototypes and group discussions
4. Cognitive technique: such as interviews

Using multiple techniques guarantee an elaborated pool of requirements. However, it involves utilization or more resources. This can result in reduced efficiency and can introduce potential contradictions. One way of guaranteeing that all requirement resources are covered is by selecting a different technique for each kind of requirement source [17]:

1. Project matrix: techniques best suited for basic characteristics of the project, e.g. project size, project complexity, project domain.
2. People matrix: techniques best suited for people involved, e.g. domain knowledge, experience.
3. Process matrix: techniques best suited for the software development process model applied, e.g. waterfall, spiral model, agile model.

An experiment conducted by Lloyd et al. [20] examined the effectiveness of various elicitations techniques. The engineers were divided into six groups, each responsible for gathering requirements and creating a Software Requirements Specification (SRS) document for a customer designated for that group. All groups used various techniques for elicitation but the groups with greater elicitation experience were better at selecting appropriate techniques. Also, the groups with more cooperation were able to produce high-quality documents.

Once the basic requirements are gathered, the analysis and design phase begins. Bugs unidentified at analysis phase stage can pose significant challenges throughout the development process. Since all phases are connected, success of one phase is connected to the success of its predecessors and successors in the software development life cycle. Traditional methodologies like waterfall are, usually, not flexible in accommodating changes during the development; which leads to unnecessary complications. Agile, on the other hand, is more open to change adjustments by using time-boxed iterations, sprints. In their study, Subhan et al. [21] explored various development methodologies with respect to requirement analysis and design. They proposed agile methodology for easy incorporation of changes during the development cycle. In a different study by Najjar et al. [22], CMMI was declared best for adjusting changes during development. In their work, Silingas et al. [23] highlighted the importance of Unified Modeling Language (UML) in capturing and demonstrating essential aspects of requirements for analysis.

Once the analysis and design phase is over, the phase of validation of requirements begin. This is where the quality and correctness of the requirements gathered is ensured [24]. Prototyping, followed by simulation or any other model-based validation was proposed a good validation strategy by Moketar et al. [25]. According to them, this sequence of validation ensured reliability, completeness, and accuracy of requirements. Reviewing requirements manually, via walkthroughs and inspections, is another way of guaranteeing completeness and correctness [26]. However, manual validations techniques need properly documented templates, tools, and protocols to conducted the review process. Also, more experienced human resources are needed for manual validation.

In short, all RE phases contribute to the quality involved in the developmental process as well as the software developed. Managing all the phases is another challenge that needs special attention throughout the RE process. Ali Altalbe [27] advocated the use of a traceability matrix to establish relationships between different requirements, facilitating better understanding and management of requirements dependencies.

3. Research methodology

This section outlines the materials and methods used to conduct the present study. It explains the research design, data collection methods, sampling strategies and analytical techniques used to investigate the research objectives.

3.1. Methodological Framework

To achieve the research objectives, two techniques were defined:

1. A systematic literature review (SLR): to explore existing knowledge and identify research gaps.
2. An industrial survey: to gather empirical insights directly from professionals in the field.

The SLR used for the study was following EBSE Technical Report by Kitchenham and Charters [28], as explained in Fig. 1. Instead of randomly searching the web for relevant data, SLR uses a defined approach of finding, sorting, analyzing and interpreting available research on a topic.

To validate the findings of the SLR, a questionnaire tool was developed. The purpose of the tool was to see if the views of software professionals matched the findings reported by the SLR. Also, if the software industry followed the same practices as reported by the SLR. The steps involved in conduction of the survey are shown in Fig. 2.

The details of both the techniques are given below:

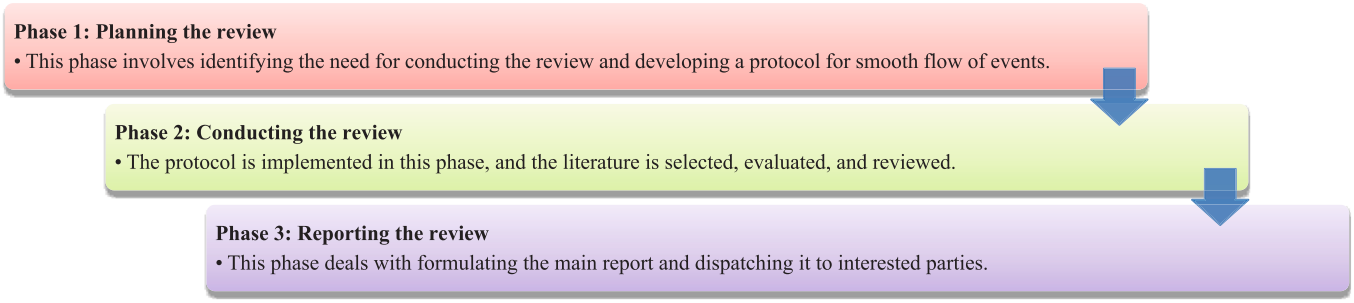

Fig. 1. Systematic literature review process.

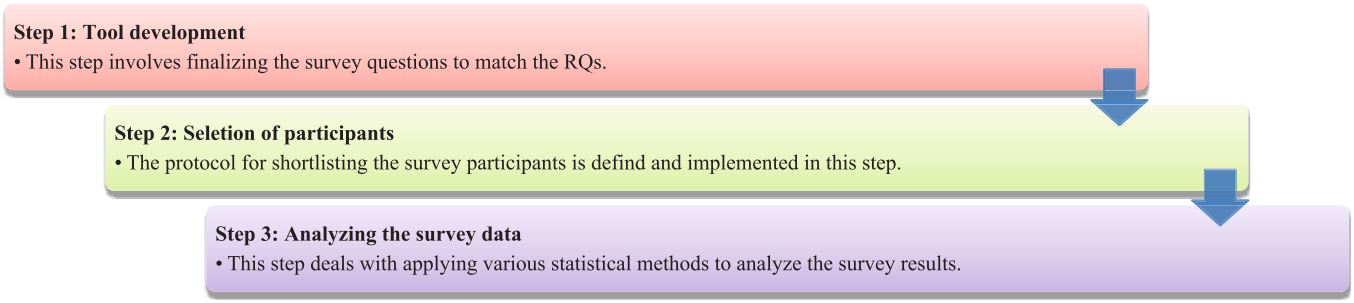

Fig. 2. Industrial survey process.

3.2. Systematic Literature Review

This section includes a detailed account of the SLR conducted for this study. It outlines the complete review protocol, including the research questions, data sources, search query, inclusion and exclusion criteria, quality evaluation criteria, and data extraction process. The goal of this review was to identify relevant studies, synthesize existing knowledge, and highlight research gaps that informed the subsequent phase of the research.

3.3. Research Questions (RQ)

The research is focused on gaining insights into practices recommended or not recommended for the quality of the RE process and any tools that can help professionals achieve better quality. So, three questions were formulated, each targeting one objective; Table 1 explains the questions and motivation behind each.

3.4. Data Sources

This study was conducted using three famous e-repositories:

1. IEEE Xplore (https://ieeexplore.ieee.org/): With a vast collection of over 5 million articles covering computer science and various engineering disciplines, this resource offers a comprehensive repository. Additionally, the most esteemed conferences globally choose to publish their proceedings through IEEE Xplore.

Table 1
The motivation behind each RQ.

RQs	Motivation
RQ1: What mistakes compromise the quality of the RE process?	Understand which practices interfere with or decrease the quality of the RE process.
RQ2: Which practices are suggested for an improved RE process?	Identify what practices are suggested to enhance the quality of the RE process.
RQ3: Which tools and techniques help to improve the RE process?	Identify the tools and techniques the researchers developed/used to improve the RE process.

2. ScienceDirect (https://www.sciencedirect.com/): Besides books and conference proceedings, it has more than 22,000 journals. Also, The Higher Education Commission (HEC) of Pakistan recognizes its indexing service, Scopus, as a valuable tool for identifying reputable journals.
3. ACM Digital Library (https://dl.acm.org/): It has an impressive collection of over 3 million published articles and is prominent in the esteemed computing community.

3.5. Search Query

The search query used was constructed using five primary keywords. The reason for the selection of each is given below:

- Requirement: The study was focused on all aspects and phases of the requirement engineering process. It also served as a stem word for:
  - o Stem words – requirement engineering, requirement engineering process, requirement gathering, requirement elicitation, requirement design, requirement analysis, requirement specification, requirement management
- Software: The study was focused explicitly on the software domain. It also served as a stem word for:
  - o Stem words – software engineering, software engineering process, software requirement engineering, software requirement engineering process
- Quality: The study was focused on enhancing the quality of the RE process. It also served as a stem word for:
  - o Stem words – quality engineering, quality enhancement, quality enhancement tools
- Problem: The study was focused on issues faced by professionals in the RE process. It also served as a stem word for:
  - o Stem words – problems, problematic
  - o Synonym added – Challenge, Issues
- Practice: the study was focused on finding practices liked or disliked by professionals for qualitative RE.
  - o Synonyms added – Technique, Process, Practices, Guidelines
  - o Stem words – practices, guidelines, techniques, processes

Hence, the complete query used was:

Requirement AND (Software AND Quality AND ((Problem OR Challenge OR Issues) OR (Practice OR Process OR Technique OR Guideline))).

3.6. Inclusion/Exclusion Criteria

The study considered the articles matching the inclusion criteria mentioned below:

- IC1: Articles having the main keyword 'Requirement' in their title.
- IC2: Articles published between January 2018 to December 2023.
- IC3: Articles published only in journals and conference proceedings.
- IC4: Articles having English as a primary language.
- IC5: Articles with an abstract matching the research objective of the study.

Any articles failing to meet the above mentioned points were excluded from the study. Duplicates were also excluded from the analysis.

3.7. Quality Evaluation (QE)

The selected articles were evaluated against the following traits, each on a pre-defined scale of 0 to 1, shown in Table 2. This was done to check an article's importance in providing an answer(s) to any or all of the RQs identified for the study. For an article to be considered valuable, it must deal with any practice or problem related to any phase of RE; i.e., the least quality score to be considered valuable was 1.

3.8. Data Extraction and Synthesis

A total of 776,904 articles were found in the three e-repositories against the search query. After applying all five selection criteria and removing duplicate items, 57 articles were left that fit the scope and objective of the study; Fig. 3 shows the complete selection process of the articles.

According to the QE criteria, articles with a total score of less than 1 were not considered valuable enough to include in the analysis. When the QE criteria were applied, the least score by any of the articles was a 2, as shown in Table 3; hence, all 57 articles were included in the analysis, proving that the selection criteria applied were well suited for the research objective. Also, about 66 % of articles scored between 3 and 4; details are shown in Fig. 4.

For RQ2 and RQ3, the selected articles were to be synthesized based on their coverage of one or more phases of the RE process, i.e., requirement elicitation, requirement analysis and design, requirement specification, requirement validation, and requirement management.

The SLR resulted in eight quality-enhancing practices that were entrusted to improve the quality of the RE process. In case of not following any of the eight practices, the results were said to be quality-diminishing. Frequency analysis was performed for the eight practices, separately as being enlisted as quality-enhancing and quality-hindering in SLR and a rank order was calculated for each quality category. An overall rank was allocated to each practice, based on its individual rank in each quality category. This was done for easier comparison with the survey findings.

Table 2
Quality evaluation criteria.

No.	Article Trait	Scale +1	+0.5	0
QE1	The article deals with a phase of the RE process	Yes	Partially	No
QE2	The article deals with a quality trend or practice related to the RE process	Yes	Partially	No
QE3	The article deals with problems associated with the qualitative RE process	Yes, with solutions	Yes, without solutions	No
QE4	The article states methodology with proof (implementation, case study, simulation)	Yes	Methodology without proof	No
QE5	The article states limitations and future directions	Yes, both	Yes, one of the two	No

3.9. Industrial Survey

This section presents the design and execution of the industrial survey conducted to validate the findings of the SLR. It outlines the process followed to develop the tool and select the participants from the software industry of Pakistan.

3.10. Tool Development

An MCQs based questionnaire was devised to validate the SLR findings. The questionnaire's primary objective was to gather professionals' insights on the RQs regarding mistakes, practices, tools, or techniques that either enhance or hinder quality. The 13 questions and corresponding close-ended answers were derived from SLR and previous studies conducted by Khalid et al. [86,87]. Approval for the questionnaire was obtained from the research's supervising committee.

3.11. Participants

The survey targeted software professionals working in various software organizations across Pakistan. Convenience sampling was used to select a sample of software professionals from Pakistani software industry. The collection of responses was facilitated through personal visits to software houses or by sending the survey via email to the participants. It's important to note that the entire process was conducted with the respondents' consent and did not involve recording any personal information.

A total of 325 questionnaires were dispatched to various software professionals around Pakistan, out of which 250 responses (response rate: 76.9 %) were received. After removing incomplete submissions, a total of 138 responses were considered for comparison; Fig. 5 shows the work experience summary of the selected respondents.

3.12. Survey Data Analysis

In order to organize the survey data, frequency analysis was used. The purpose was to see which practices were causing quality issues with the software professionals and which practices were majorly deployed by software organizations to ensure quality. Since each question was offering a mixture of options against the identified quality practices, a rank analysis was conducted for each question, based on the frequencies and an average rank was computed to see which practices were most common in the software organizations.

4. Findings

This section summarizes the findings of the SLR and the industrial survey.

4.1. Findings from SLR

An SLR was conducted to see what practices were reported by researchers to ensure quality in the SRE process. Below are the findings against the three RQs from the shortlisted articles.

RQ1: What mistakes compromise the quality of the RE process??

This first research question focused on mistakes compromising quality the RE phases. The mistakes highlighted by the shortlisted articles were divided into procedural and managerial level mistakes to see

Fig. 3. Identification of studies.

how often each was reported; shown in Table 4.

RQ2: Which practices are suggested for an improved RE process?

The second question focused on practices or techniques reported by literature that could practically improve the quality of the RE phases and positively impact the overall quality of the RE process. To better understand the practices, the findings were analyzed keeping in mind the explicit phases of RE, i.e., elicitation, analysis and design, specification, and management, in Table 5. No practices were found for the validation phase.

RQ3: Which tools help to improve the RE process?

The third research question focused on any tools or models developed or used by researchers to help aid the quality of the RE process. To understand the role of each reported tool or model, the findings were explored explicitly for each of the five RE phases in Table 6.

4.2. Summary of Quality-Enhancing Practices from SLR

Around 50 % of requirements fail to retain their original form before reaching the development phase [55]. Thus, the cycle of RE is very crucial and complex. Based on the findings yielded by the shortlisted articles of SLR, the following can be said about quality-ensuring practices for the RE process:

11. Template usage: natural language causes misunderstanding between team members or customers while eliciting or specifying requirements. Using pre-defined templates to ensure precise terminologies and language uniformity reduces the chances of missing important information. These templates must be updated as the development progresses.
21. Requirements' clarity: vague or unclear requirements cause issues leading to project failure. Also, exploring the target market, similar pre-existing systems, any dependent components, and all possible new sources can reduce the chances of missing on important requirements. Involving seniors or experts to gain a better understanding can also ensure the correctness of requirements.
31. Analysis accuracy: improper or wrong analysis of requirements, in terms of cost and time estimates or requirements' dependency, creates problems later on.
41. Requirements' priority: an end-product that does not provide any business value is considered a failure. Hence, the requirements must always be prioritized as per the customer or market needs. Prioritization also helps in understanding requirements' dependency.
51. Validation of quality: a qualitative product can only come from a qualitative process. Hence, quality checks should be applied from the very beginning to avoid late detection of errors and other catastrophic failures. Late or missed requirement validation hinders progress.
61. Tracking of changes: change requests generated can alter the existing dependencies and architectures, so they must be properly tracked in order to avoid misfortunes. Also, the templates must be updated to help in tracking.
71. Customer involvement: customers must not only be involved while gathering the requirements, but they must also be updated about every milestone achieved.
81. Team coordination: train and synchronize the team about relevant documents and development processes.

The shortlisted articles have discussed the above eight practices as quality-ensuring, if followed. Also, the consequences of not following them have been explored and declared as quality-hindering in the articles. Table 7 shows the frequency and rank of articles exploring the eight practices as quality-enhancing and quality-hindering in the SLR. The overall rank of each practice was based on the total frequency.

4.3. Findings from Industrial Survey

This section summarizes the findings of the industrial survey in light of the eight practices highlighted by SLR. The 1st subsection evaluates the industrial survey with respect to quality hindering practices, the 2nd subsection evaluates the survey with respect of quality enhancing practices, and the third subsection summarizes the practices against all

Table 3
QE of 57 selected articles.

Ref.	QE1	QE2	QE3	QE4	QE5	Total
[29]	1	1	1	1	0.5	4.5
[30]	1	0	0.5	0	0.5	2
[31]	0	0	0.5	1	1	2.5
[32]	1	1	1	1	0.5	4.5
[33]	1	0	0	1	1	3
[34]	1	0	1	1	0	3
[35]	1	0	1	1	1	4
[36]	0.5	0.5	1	1	0.5	3.5
[37]	1	1	1	1	0.5	4.5
[38]	1	1	1	1	1	5
[39]	1	1	1	1	0.5	4.5
[40]	0.5	0.5	1	1	1	4
[41]	1	0.5	1	1	0.5	4
[42]	1	0	1	1	0.5	3.5
[43]	1	1	0.5	1	0.5	4
[44]	1	0.5	0	0	0.5	2
[45]	1	1	1	1	0	4
[46]	1	0	1	1	0.5	3.5
[47]	0.5	0	1	1	0.5	3
[48]	1	0	0.5	0.5	0	2
[49]	1	0	1	1	0.5	3.5
[50]	0.5	0	1	1	0	2.5
[51]	0.5	0	1	1	1	3.5
[52]	1	1	1	1	0.5	4.5
[53]	0.5	0	1	1	1	3.5
[54]	1	0.5	1	1	0.5	4
[55]	1	0.5	1	1	0.5	4
[56]	0.5	0	0.5	1	1	3
[57]	0.5	0.5	1	1	1	4
[58]	0.5	0	1	1	0.5	3
[59]	0.5	0.5	1	1	0	3
[60]	1	0.5	1	1	0	3.5
[61]	0.5	0.5	1	1	0.5	3.5
[62]	0.5	1	1	1	0	3.5
[63]	0.5	0	1	1	0.5	3
[64]	0.5	0.5	1	1	1	4
[65]	1	0.5	1	1	1	4.5
[66]	1	1	0	0	0.5	2.5
[67]	1	0	0.5	1	0.5	3
[68]	1	1	0.5	0	1	3.5
[69]	1	0	0.5	0.5	0	2
[70]	0	0	0	1	1	2
[71]	1	0	1	1	1	4
[72]	0.5	0	1	0	0.5	2
[73]	0.5	0	0	1	0.5	2
[74]	1	0	1	1	0	3
[75]	1	0.5	1	1	0	3.5
[76]	0.5	1	0	1	0.5	3
[77]	0.5	0.5	0.5	1	0.5	3
[78]	1	1	1	0	0.5	3.5
[79]	0	0.5	1	1	1	3.5
[80]	0.5	0.5	0	1	1	3
[81]	1	0.5	1	1	1	4.5
[82]	1	0.5	1	1	0.5	4
[83]	0	0.5	0.5	0.5	0.5	2
[84]	1	0.5	0	0.5	0.5	2.5
[85]	1	0	0.5	1	1	3.5

12 survey questions, based on the frequency of occurrences, and allocates a rank to each practice.

4.4. Quality-Hindering Practices

The first set of questions targeted practices or reasons which negatively impacted the RE process, i.e. quality-hindering practices. Each question was followed by a predefined list of options derived from the findings of the systematic literature review. Participants were asked to select the options they had experienced or observed in their professional context. Frequency analysis was performed on the responses to identify commonly reported issues and trends. This helped in quantifying the prevalence of each practice across different organizations.

When discussing problems with the elicitation phase, shown in Fig. 6, most professionals (n = 75) thought that gathering clear and unambiguous requirements was the most challenging part. The second highest issue was estimating time and budget (n = 55), followed by an uninterested customer (n = 46). Professionals were also experiencing the remaining problems on a vast scale. But the top three voted were following the literature where inconsistent requirements and wrong estimates were being highlighted as quality diminishing causes. In other words, P2, P3, P7, P8, and P4 were rated as existing software industry problems, respectively. This confirmed the presence of 5 out of the 8 practices in the quality-hindrance capacity.

According to the survey results depicted in Fig. 7, ambiguous requirements were identified as the primary cause of an extended analysis phase (n = 76); following closely behind was the challenge of prioritizing requirements (n = 27). The top three reasons from Fig. 5 can result in ambiguous requirements, conforming to the former findings. Here, P2, P4, P8 and P3 were ranked top to bottom as being the quality-hindrance practices during the analysis phase in the software industry.

Fig. 8 links two separate questions about which mistake creates the most trouble at the project completion phase and who is to blame for it. The primary cause of inadequate analysis was mishandled change requests (n = 70). Moreover, an inexperienced software professional was voted as the major cause (n = 92) of any mishaps occurring at project completion. This highlighted P6, P1, P3, P2, and P5 being the quality-hindering practices at the project completion phase for one question and P7 and P8 being the quality-hindering practice under the party responsible for the issue caused.

The findings of Fig. 9 connect with those of Fig. 6, with poor requirements being the most problematic attribute in a project and causing project failure at any phase, from the elicitation to the final delivery. This indicates that professionals must ensure their initial understanding of the customer's request and the market needs. Also, they must document it appropriately to avoid losing its true essence during the development cycle. This marks P2, P7, P3, P1, and P8 as the cause of project failure in the quality-hindering capacity.

4.5. Quality-Enhancing Practices

For the second set of questions, the focused was shifted to practices followed in the industry that improve the quality of the RE process and

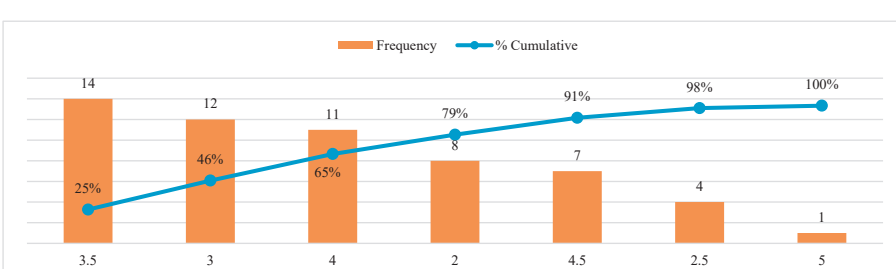

Fig. 4. Quality evaluation score of shortlisted articles.

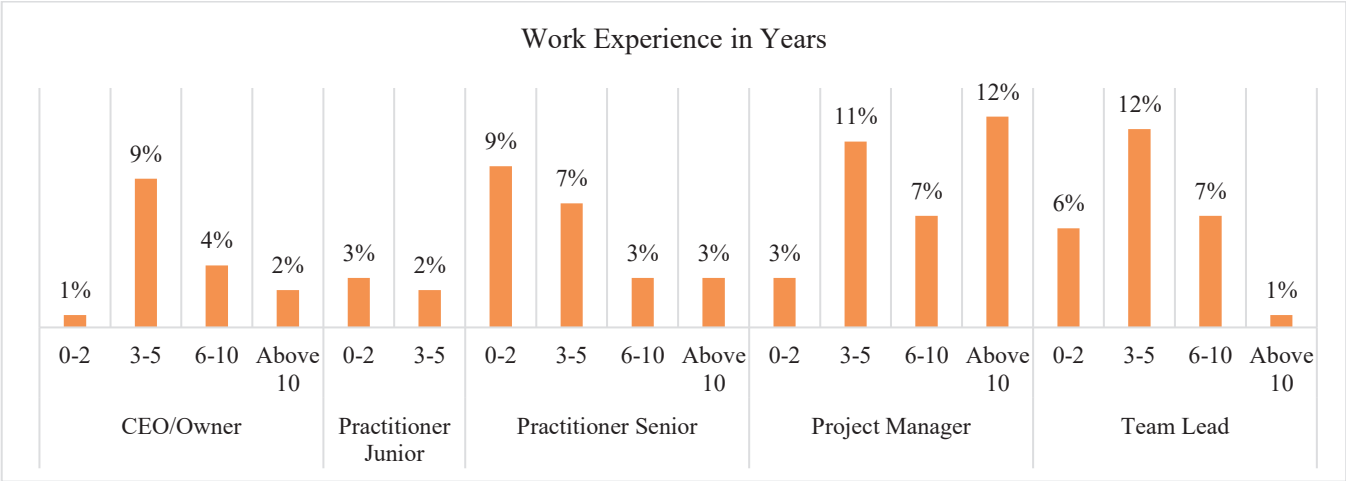

Fig. 5. Respondents' work experience summary.

Table 4
Quality-hindering mistakes from SLR.

	Mistakes	Article No.
Procedural Mistakes	Team members and customers having no or distinct understanding of the requirements	[29 30 35 37 52 56 65 66 79 85]
	Misunderstood emergent properties	[31 42 51]
	Misunderstood crosscutting concerns	[35 36 77]
	Late detection of a defect or error; leads to more rework or catastrophic results	[29 37 40 45 56 55]
	No defined templates for recording requirements, resulting in missing or vague requirements	[29 37 38 39 40 47 52 55 67 75 76 78]
	Inconsistent or imprecise usage of language, leading to unnecessary complexity	[29 39 41 43 55 62 67 71]
	Changes in requirements, affecting project time and cost negatively	[63 64 67 68 69]
	Wrongly prioritized requirements	[24 37 59 60 61 74]
Managerial mistakes	No synchronization between the ongoing development/updates and documents, creating confusion	[29 63 64 85]
	Globally distributed teams, causing inconsistencies in understanding	[59 67 68]
	Social conflicts or cultural differences in the working environment, having a bad influence on the quality of the requirement and/or development process	[39 66 71]

the product, i.e. quality-enhancing practices. Each question was directed at gaining insights on at least one practice identified from the literature. The options were derived from previous literature. Furthermore, specific questions were designed to link RQ3 with the findings about techniques used by professionals to ensure certain practices.

From Fig. 10, it is evident that software professionals strongly preferred the analysis of existing systems or documents as an elicitation technique (n = 82). After that, interviews (n = 65) were a preferred technique for requirement elicitation. The remaining three options were also being used by many. In light of these responses, it was clear that software professionals were using existing systems to understand project domains before formally starting the elicitation process, mainly done by conducting interviews. Also, the presence of customer was necessary, and hence, his absence causing issues (Fig. 6) was proved. The complete focus of this question was on P2 as it explored various techniques used in the industry to gather clear and unambiguous requirements.

As shown in Fig. 11, the customer was updated about the feasibility and analysis updates of the requirements (n = 93). This question was verifying the P7 indirectly, by asking if the professionals shared the updates and understanding with the customer.

Fig. 12 shows that the practice of requirements prioritization is rigorously followed in the industry (n = 90). Although the question was directed at focus of prioritization, the objective was to confirm if the software professionals even considered prioritization. This question was directed at P4; besides a few (n = 6), the professionals prioritized requirements for one reason or the other.

As the literature mentioned, the formal use of templates and consistent terminologies as a good practice in P1, the survey question directly asked the professionals about their requirement recording practices. As shown by Fig. 13, most professionals (n = 60) were using semi-formal methods to record requirements. However, a fairly large number (n = 56) of professionals was doing exactly the opposite of what was recommended, i.e., recording requirements in the native language. As mentioned in previous findings, this might also explain the experiences of professionals with inadequate and ambiguous requirements.

Fig. 14, where the most favored method of validation was directly asking the customer, tells why the unavailability of the customer was an issue at the elicitation phase, as per Fig. 5. Also, these findings conform with the customer involvement practice of Fig. 11, as to ask the customer to validate quality, the customer must be kept updated to the development progress. Since the customer was frequently involved for quality approval, he could not be blamed for inadequate analysis (Fig. 8). This question was focused mainly on P3 and P5 but was also covering a part of P7 and P2 as sources of validation.

Changes can greatly impact the development cycle since requirements need proper time and attention for the project to succeed. According to Fig. 15, software professionals (n = 95), when catering to change requests, asked for more time or budget for the cycle to not be affected negatively. Here, the professionals' views favored the literature findings of P6, and they reported managing change requests by timelines and budget; since both directly impact the customer, a reasonable customer would think carefully before asking for changes in the requirements.

The final two questions were focused on generic views about development. The first question inquired if the quality of the final product was given due importance, as compared to timelines, as shown in Fig. 16. This also favors the professionals' choice of accommodating changes at the cost of more time and budget (Fig. 15). Under no circumstances the software professionals (n = 95) were compromising on quality. Besides P6, this also proves the importance of formulating practices and methods that enhance the quality of the overall SRE cycle.

The last question was a general view of the RE phases. Since the whole RE process resides on the quality of requirements gathered, the

Table 5
Quality-enhancing practices from SLR.

Requirement Elicitation	Requirement Analysis and Design	Requirement Specification	Requirement Management
• Involving a domain expert and the customer helps a business analyst understand requirements more precisely [29]. • Developing an appropriate user persona and selecting the correct elicitation technique for requirement gathering [30]. • Studying the interdependency between software and hardware components for developing a product [31]. • Ensuring effective collaboration between team and stakeholders [75]. • Redesigning communication processes for avoiding elicitation errors [79].	• Including a whole group of domain experts reach a mutual decision regarding requirements [32]. • Having an experienced team deduce models from requirements and make decisions improves the quality of results generated [33]. • Comparing various modeling techniques to represent requirements results in successful projects that meet business needs and satisfy stakeholders [34]. • Using analytic network process (ANP) methods to analyze the requirements' dependency on each other to improve decision-making [35]. • Using mining tools for semantic analysis of requirements for early identification of crosscutting concerns [36]. • Improving existing modeling languages to accommodate AI based software [80].	• Using formal methods to precisely record requirements [37]. • Using proper sentence structure for defining the requirements for improved document quality. [29]. • Using boilerplates for standardized requirement specification by [39 40]. • Involving a third party to identify incomplete or vague requirements [39].	• Using quality function deployment tools to focus on customers' end product needs [35]. • Using business process models for quality-induced requirements [41]. • Using goal-oriented RE to improve the quality of the end product [42]. • Using ontology-driven RE for better quality [43]. • Using machine learning techniques to help prioritize a large number of requirements efficiently [44 57]. • Using supervised learning to categorize quality requirements for better understanding [32]. • Using POS tags to categorize requirements as faulty vs. non-faulty or as vague or clear [37,45]. • Using requirement patterns (RPs) and test patterns (TPs) ensure quality, along with increased productivity [65]. • Formulating a taxonomy of human errors for each phase of RE to help reduce cost and improve quality [79].

most critical and difficult phase is, no doubt, the elicitation phase, as per the professionals answered in Fig. 17. If the requirements are not properly gathered, none of the following phases can add quality to the end product. Also, the initial phases of the RE cycle, i.e., requirement elicitation, requirement analysis, and requirement specification, take 70 % of the total problematic space in the SRE process. This is because the whole project depends on the quality of the requirements gathered. But since the organizations are not using formal templates to record requirements, these phases are bound to be the most affected parts.

In short, the professionals' views were in accordance with the literature findings, as most organizations were careful about gathering their initial set of requirements and about keeping the customer updated and in the loop of development. Similarly, the delay in the analysis was most commonly reported to be due to poor quality of requirements, and the team skills were responsible for this, just as the literature predicted and warned against in P8.

4.6. Comparison Summary of Practices from Survey

Views of the software professionals confirm the findings from the literature in many regards. Similarly, it also highlights some quality-diminishing practices being followed and the issues being caused due to them, which also confirms why the literature has warned against such practices. Table 8 gives a distribution of the eight practices in all 12 questions of the industrial survey; the last question was generic and was not directed at any practice and is not included in the table.

5. Comparison between SLR and industrial survey

To get a clearer understanding of the current industry practices in comparison with the literature findings, the occurrence frequencies of both were compared against each other. Table 9 and Fig. 18 and Fig. 19 give a rank-based comparison between practices from SLR and their confirmation by the software professional via an industrial survey, the ranks are based on percentage frequency calculated using Table 4 and Table 5.

According to the average rank, P2 was the most important practice, whereas, P3, P4, P5, and P7 were equally ranked. However, if we look at separate ranks of each practice, P1 was ranked at 2 in the SLR, whereas, it was ranked last by the professionals. Similarly, P3 was the top ranked practice in the SLR and it was ranked at 7th position by the industrial survey. This difference indicated that there exists a gap between literature and industry. The real-life experiences of software professionals were not being focused upon enough by the research community. This supports the fact of the whole RE cycle residing on how well defined or understood the requirements are.

To further evaluate the relation between the practices from the SLR and the industrial survey, a paired *t*-test and a goodness-of-fit test via regression analysis were run on the average frequencies from both groups. Tables 10 and 11 show the paired differences and statistics on both ranks. The correlation of −0.412 shows a moderately negative relationship between the industry and SLR, while the significance value of 0.254 suggests the relationship to be of no significance. The t-value of 1.244, with a significance of 0.254, also suggests that the mean difference between the pairs is not statistically significant.

According to the goodness-of-fit test, the R^2 value of 0.1952 also indicates a very weak relationship between SLR and Industry, with only 19.52 % of the variance in industrial rank explained by SLR rank. Moreover, the adjusted R^2 value of 0.0611 also indicate a weak fit, with a standard error of 2.369963025. The ANOVA (Table 12) and coefficient (Table 13) results also show that the SLR rank is not a significant predictor of industrial rank.

Furthermore, the residual and the probability output (Table 14) shows that the model may not capture the full complexity of the relationship between SLR and industry rank. Despite there are no outliers in the standardized residuals, the variance indicates that industry rank predictions are not consistently accurate across different percentiles.

Hence, the tests validate that the RE practices mentioned in the literature are not reflected in the industry. This means that the research community needs to take a closer look at industry trends and propose

Table 6
Quality-enhancing tools from SLR.

Requirement Elicitation	Requirement Analysis and Design	Requirement Specification	Requirement Validation	Requirement Management
• Using recommender systems [46] e.g. CoRM [47]. • Extracting requirements from existing documents and product forums [32,48,49], e.g. BERT-CNN [82].	• Analyzing existing software components and their dependency on each other to model collaborations for a System of Systems, e.g. ADORE [42]. • Using user centered designs to understand the link between requirement models, e.g. Agile-UCD [50]. • Performing semantic analysis at word and sentence level to construct a similarity network to identify variability in features of a software product line [51], e.g. TEMPOR [78]. • Acquiring testable non-functional requirements from goal-driven product lines, e.g. AUCMT [75]. • Using petri-nets based tools (Stanford CoreNLP and PIPE2) to automatically analyze use cases and scenarios [81]. • Using machine learning to segregate good and bad quality requirements [83].	• Using a properly developed SRS document [38,39,40 52], e.g. Automated Requirements Measurement (ARM) and Quality Analyzer of Requirements Specifications (QuARS) [29].	• RESSM to help software organizations validate their security requirements [53]. • Using virtual prototyping to reduce the efforts involved in the manual validation [55]. • Goal-Question-Metrics technique to assess quality of requirements via requirement metrics [84]. • REsHFv, using software visualization specifications, to validate requirements [54]. • Using Human Error Taxonomy to identify and reduce human errors [56].	• Formulating proper methods to prioritize requirements [58 59], e.g. AHP [60,61,74], EVOPM [57], and RADIATION [77]. • ARTM to trace unimplemented requirements and track changes made to requirements [63]. • GuideGen to automate guidance about updating acceptance tests when a linked requirement changes [64]. • SoPaMM to reuse RP in automatically generating TP [65]. • OWL ontology for requirement traceability [62].

Table 7
RE Practices as quality-enhancing and quality-hindering in SLR (n = 57).

ID	RE Practices	Discussed as Quality-Hindering Practice			Discussed as Quality-Enhancing Practice			Total Frequency	Overall Rank
		Frequency	%	Rank	Frequency	%	Rank		
P1	Template usage	14	25	1	13	23	3	27	2
P2	Requirements' clarity	12	21	3	7	12	5	19	4
P3	Analysis accuracy	13	23	2	22	39	1	35	1
P4	Requirements' priority	7	12	5	16	28	2	23	3
P5	Validation of quality	6	11	6	10	18	4	16	5
P6	Tracking of changes	5	9	8	6	11	6	11	7
P7	Customer involvement	9	16	4	4	7	8	13	6
P8	Team coordination	6	11	6	5	9	7	11	7

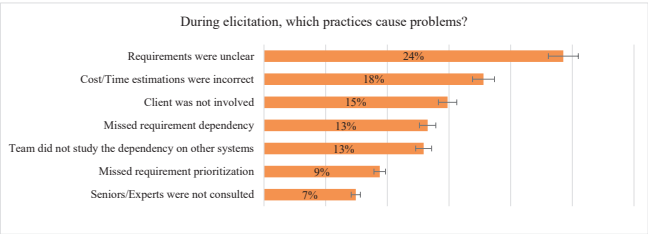

Fig. 6. Practices causing problems in the elicitation phase.

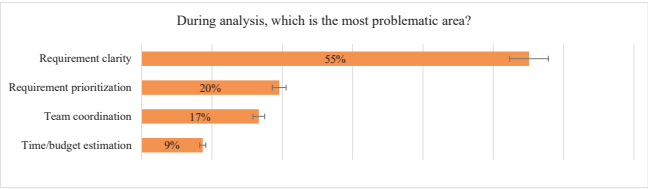

Fig. 7. Reasons for prolonged analysis.

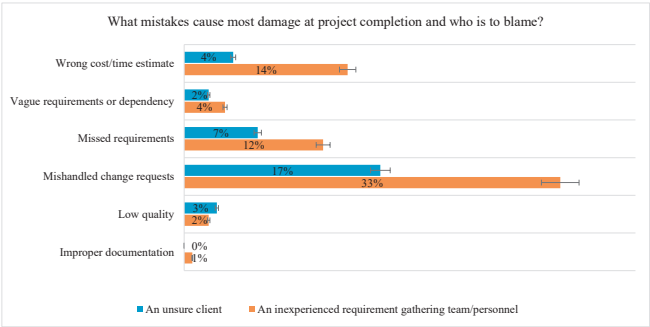

Fig. 8. Reasons for inadequate analysis.

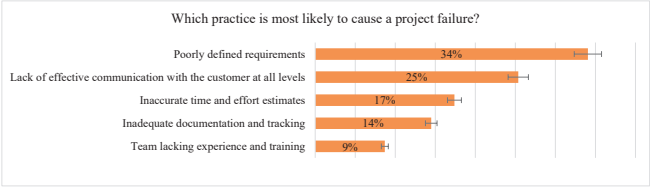

Fig. 9. Reasons for project failure.

suggestions suitable for practices followed. This gap between the literature recommendations and industry practices, indicate that finding new ways to enhance quality and ensure quality in existing SRE processes is of utmost importance. This needs collaboration from both the

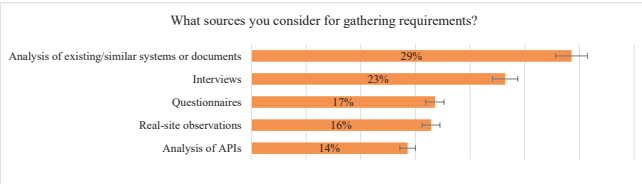

Fig. 10. Sources of requirement elicitation.

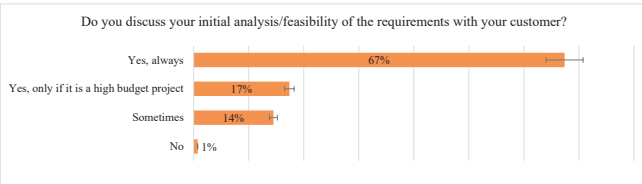

Fig. 11. Involvement of customer.

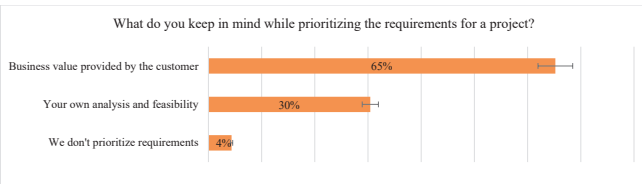

Fig. 12. Prioritization of requirements.

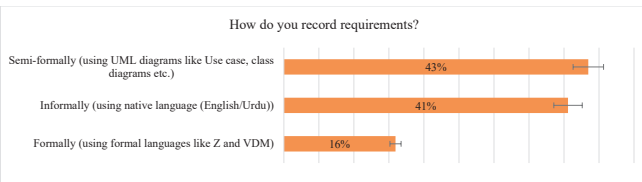

Fig. 13. Practice of recording requirements.

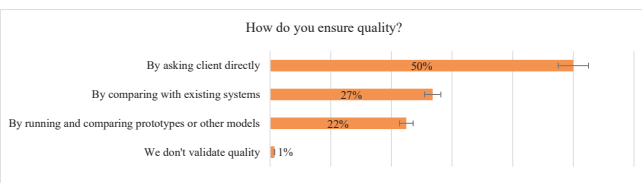

Fig. 14. Methods for quality validation of requirements.

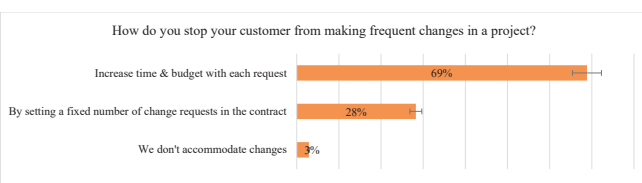

Fig. 15. Practice of controlling change requests.

industry and the research community.

6. Conclusion

A success of a software product depends on the quality of its functionality, making RE the most crucial and success-guaranteeing part of the software development lifecycle. Poorly managed requirements can

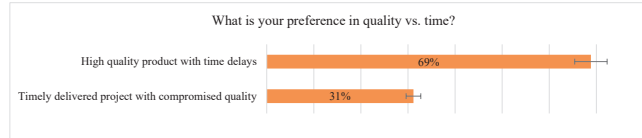

Fig. 16. Time vs. Quality.

easily lead to project or product failure. Researchers have provided many insights about practices to improve the quality of requirements and the RE process. This study compares the literature presented on practices that ensure quality in SRE with the trends in the software industry. Three e-repositories were selected to search for relevant literature published between 2018 and 2023 in conferences or journals. Out of the 776,904 articles found, 57 were selected for analysis against the defined search criteria.

The analysis of the SLR resulted in highlighting eight practices that were linked to quality: template usage, requirements' clarity, analysis accuracy, requirements' priority, validation of quality, tracking of changes, customer involvement, and team coordination. The findings suggest that adhering to these practices significantly contributes to producing higher-quality software. Conversely, neglecting them often leads to diminished quality and increased risk of project failure.

Based on the eight practices, a survey questionnaire was designed to gain insights from the software professionals on the quality-related practices and to validate them in return. The survey results were largely consistent with the existing research, confirming that the absence of these eight practices in the industry often led to the same issues as highlighted in the literature. Among the most critical challenges identified were unclear and ambiguous requirements, which were validated to be the top most problematic area during requirement gathering and analysis, and the top cause of project failure. The findings also indicated that the responsibility for faulty requirements more often lay with the team responsible for gathering them, rather than with uncertain customers. Quality was better ensured by engaging the customers directly and keeping them updated on the development progress. Overall, the survey strongly supported the relevance and necessity of all eight RE practices for enhancing both the quality of the software process and the resulting products.

The comparison between the practices identified through the SLR and those observed in the software industry revealed a noticeable gap between the challenges faced by the professionals and what areas were emphasized by the researchers. Statistical analysis supported this observation, with a t-value of 1.244 indicating that the mean difference between the SLR and industrial findings was not statistically significant. Additionally, an R^2 value of 0.1952 suggested a very weak correlation between the two sets of findings. Overall, the statistical analysis revealed that the quality practices for RE, as mentioned in the literature, were not consistently reflected in the software industry. Nevertheless, requirements' clarity emerged as the top ranked quality-ensuring practice across both the academic and industrial contexts.

This highlights the importance of conducting more quality-themed research in the software industry to improve the development cycle in terms of quality. Also, such studies give an insight to professionals about the latest trends and techniques being recommended by the literature for quality improvement. In order to achieve the mutual goal of better quality, the research community and the industry need to work together to explore and experiment with latest trends and practices.

7. Limitations and future work

7.1. Selection Bias

- The articles selected for this study were taken from conferences or journals of three e-repositories, published between 2018 and 2023;

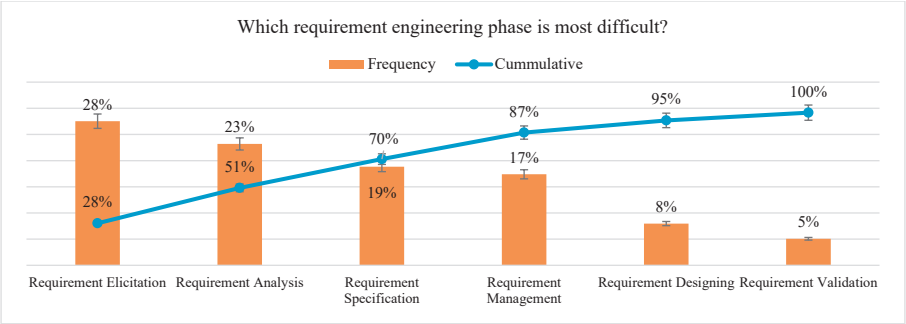

Fig. 17. The difficulty level of each RE phase.

Table 8
Distribution of practices from SLR in the industrial survey.

		P1	P2	P3	P4	P5	P6	P7	P8
Q1	Frequency	0	138	96	29	0	0	46	0
	Rank	5	1	2	4	5	5	3	5
Q2	Frequency	0	76	12	27	0	0	0	23
	Rank	5	1	4	2	5	5	5	3
Q3	Frequency	1	34	34	0	7	70	0	0
	Rank	5	2	2	6	4	1	6	6
Q4	Frequency	0	0	0	0	0	0	46	92
	Rank	3	3	3	3	3	3	2	1
Q5	Frequency	20	47	24	0	0	20	35	12
	Rank	4	1	3	7	7	4	2	6
Q6	Frequency	0	280	0	0	0	0	0	0
	Rank	2	1	2	2	2	2	2	2
Q7	Frequency	0	0	0	0	0	0	117	0
	Rank	2	2	2	2	2	2	1	2
Q8	Frequency	0	0	0	132	0	0	0	0
	Rank	2	2	2	1	2	2	2	2
Q9	Frequency	82	0	0	0	0	0	0	0
	Rank	1	2	2	2	2	2	2	2
Q10	Frequency	0	37	0	0	137	0	69	0
	Rank	4	3	4	4	1	4	2	4
Q11	Frequency	0	0	0	0	0	138	0	0
	Rank	2	2	2	2	2	1	2	2
Q12	Frequency	0	0	0	0	95	0	0	0
	Rank	2	2	2	2	1	2	2	2
Total Frequency		103	612	166	188	239	228	313	127
Overall Rank		8	1	6	5	3	4	2	7

Table 9
Occurrence of practices in SLR and Industry.

ID	Practices	Occurrence in SLR (n = 57)		Occurrence in Industry (n = 138)		Average Rank
		%	Rank	%	Rank	
P1	Template usage	47	2	25	8	5
P2	Requirements' clarity	33	4	74	1	3
P3	Analysis accuracy	61	1	30	7	4
P4	Requirements' priority	40	3	45	4	4
P5	Validation of quality	28	5	58	2	4
P6	Tracking of changes	19	7	55	3	5
P7	Customer involvement	23	6	45	4	5
P8	Team coordination	19	7	31	6	7

there can be relevant articles in other repositories or in the same repositories (published beyond the decided timeline).

- The validation was done on a convenience-based sample, and the results might vary if applied to a different sample. However, the comparison will remain the same, and the practices reported will, in any case, produce qualitative results in the SRE process.

The cited literature and selected software professionals were mainly from the traditional side of software development, where well-defined requirements are focused upon. However, the software development spectrum is much more complex. In many modern, agile, or startup contexts, the requirements may evolve or be unclear to the client or even the market at the beginning. The complete picture or a broader insight into literature and industry addressing the heterogeneous nature of software development can be done as future work.

Similarly, the survey was conducted within a single country, hence, the findings may not be globally generalizable. Different regions may have unique practices and challenges in software development, which can be a part of a future study. A more international survey would provide a more nuanced understanding of the industry's relationship with academic practices.

Moreover, in the future, the findings of this study can be used to analyze the practices being followed in some other samples from the software industry. Also, a more detailed analysis can be performed for dedicated phases of the SRE process to better understand each phase aspect.

8. Ethics Declaration

a) Competing Interests: All authors whose names appear on the submission approved this version to be published with no competing interests.

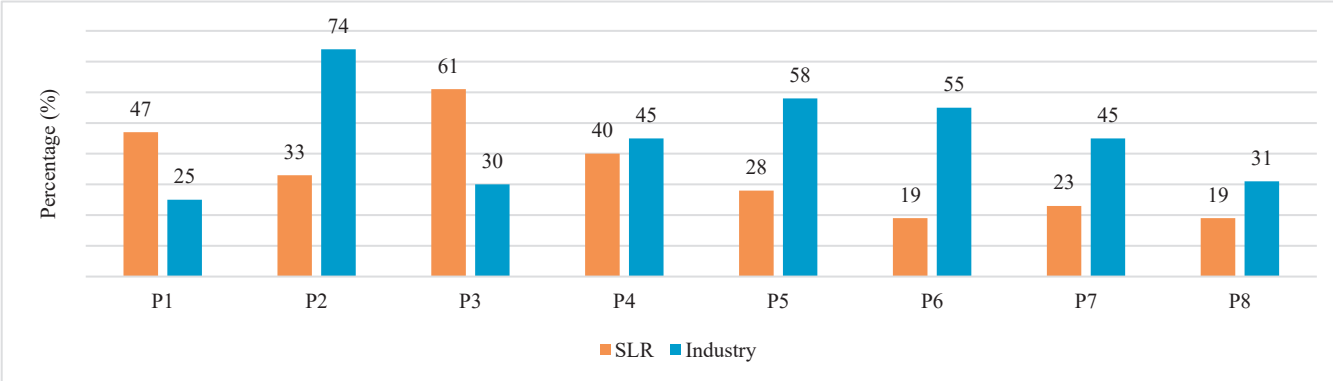

Fig. 18. Occurrence of practices in SLR and industry.

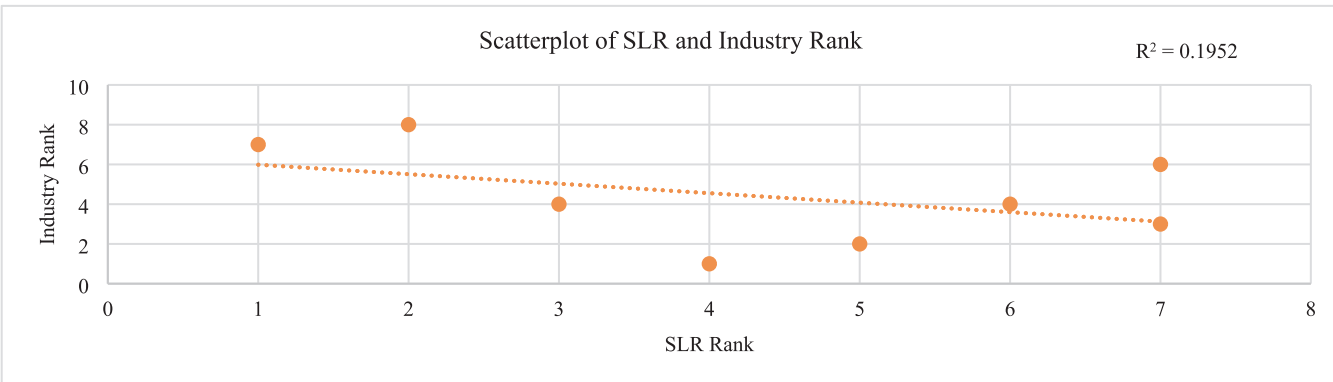

Fig. 19. Scatterplot of the mistakes' rankings from SLR and the industry.

Table 10
Paired differences between SLR and industry.

	Mean	Std. Deviation	Std. Error Mean	95 % Confidence Interval of the Difference Lower	Upper	t	df	Sig. (2-tailed)
Industry − SLR	11.62500	26.44097	9.34830	−10.48021	33.73021	1.244	7	0.254

Table 11
Paired statistics and correlation.

	Mean	N	Std. Deviation	Std. Error Mean	Correlation	Sig.
SLR	33.7500	8	14.83962	5.24660	−0.412	0.310
Industry	45.3750	8	16.60411	5.87044		

Table 12
ANOVA for goodness-of-fit.

	df	SS	MS	F	Significance F
Regression	1	8.174652	8.174652	1.455413	0.273066
Residual	6	33.70035	5.616725		
Total	7	41.875			

Table 14
Residual and probability output.

RESIDUAL OUTPUT				PROBABILITY OUTPUT	
Observation	*Predicted Industry Rank*	*Residuals*	*Standard Residuals*	*Percentile*	*Industry Rank*
1	5.508711	2.491289	1.135419	6.25	1
2	4.554007	−3.55401	−1.61976	18.75	2
3	5.986063	1.013937	0.462107	31.25	3
4	5.031359	−1.03136	−0.47005	43.75	4
5	4.076655	−2.07666	−0.94645	56.25	4
6	3.121951	−0.12195	−0.05558	68.75	6
7	3.599303	0.400697	0.18262	81.25	7
8	3.121951	2.878049	1.311686	93.75	8

Table 13
Coefficients of the simple linear regression.

	Coefficients	Standard Error	t Stat	P-value	Lower 95 %	Upper 95 %	Lower 95.0 %	Upper 95.0 %
Intercept	6.463415	1.923231	3.360706	0.015216	1.757437	11.16939	1.757437	11.16939
SLR Rank	−0.47735	0.395681	−1.2064	0.273066	−1.44555	0.490846	−1.44555	0.490846

b) Ethical Approval: The study was approved by the research's supervising committee from the Department of Computer and Software Engineering, College of Electrical and Mechanical Engineering, National University of Sciences and Technology, Islamabad, Pakistan.
c) Consent to Participate: The study did not collect any sensitive or personal information. Moreover, each participant was approached and/or contacted individually and their participation in the study was completely consensual.

CRediT authorship contribution statement

Sadia Khalid: Writing – original draft, Data curation, Conceptualization. **Uzair Rasheed:** Methodology, Formal analysis. **Uzair Khaleeq uz Zaman:** Resources, Project administration, Investigation. **Majed Alfayad:** Writing – review & editing, Formal analysis. **Mohammed Assiri:** Visualization, Validation. **Wasi Haider Butt:** Writing – review & editing, Supervision. **Mamoona Humayun:** Writing – review & editing, Supervision, Conceptualization. **Mahmood Niazi:** Visualization, Resources, Project administration.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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